# Discrete Math W5

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October 20, 2026

ETH Zurich

# Hello

I am Diego

3rd Semester CS Student

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# Group Chat



# Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

# Feedback

# Theory Recap

# Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

## 3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

# Relations

**Definition 3.9**

A *(binary) relation* $\rho$ from a set $A$ to a set $B$ (also called an $(A, B)$-relation) is a subset of $A \times B$. If $A = B$, then $\rho$ is called a relation *on* $A$.

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For any set $A$, the *identity relation* on $A$, denoted $\mathrm{id}_A$ (or simply $\mathrm{id}$), is the relation

$$\mathrm{id}_A = \{(a, a) \mid a \in A\}.$$

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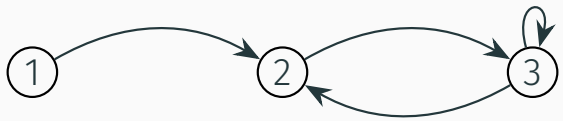
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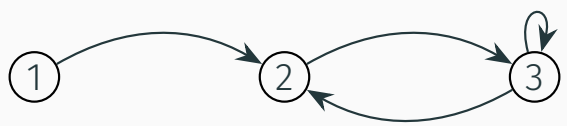
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Relations are sets, so all set operations apply ($\rho \cup \sigma$, $\rho \cap \sigma$, $\rho \subseteq A \times B$). On a finite set $A$ there are $2^{|A|^2}$ relations on $A$.

# Inverse and Composition

**Definition 3.11**

The *inverse* of a relation $\rho$ from $A$ to $B$ is the relation $\widehat{\rho}$ from $B$ to $A$ defined by

$$\widehat{\rho} \stackrel{\text{def}}{=} \{(b, a) \mid (a, b) \in \rho\}.$$

**Definition 3.12**

Let $\rho$ be a relation from $A$ to $B$ and let $\sigma$ be a relation from $B$ to $C$. Then the *composition* of $\rho$ and $\sigma$, denoted $\rho \circ \sigma$ (or also $\rho\sigma$), is the relation from $A$ to $C$ defined by

$$\rho \circ \sigma \stackrel{\text{def}}{=} \{(a, c) \mid \exists b\, ((a, b) \in \rho \wedge (b, c) \in \sigma)\}.$$

The $n$-fold composition of a relation $\rho$ on a set $A$ with itself is denoted $\rho^n$.

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**Example:** $\rho = \{(1, 2), (2, 3)\}$, $\sigma = \{(2, 3), (3, 1)\}$ on $\{1, 2, 3\}$:

$$\rho \circ \sigma = \{(1, 3), (2, 1)\} \qquad \sigma \circ \rho = \{(3, 2)\}$$

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so composition of relations is *not commutative*.

"is child of" $\circ$ "is brother of" $=$ "is nephew of".

# Socks and Shoes

**Lemma 3.7**

The composition of relations is *associative*, i.e., we have $\rho \circ (\sigma \circ \phi) = (\rho \circ \sigma) \circ \phi$.

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In the morning you put on *socks* ($\rho$), then *shoes* ($\sigma$) $\implies$ $\rho\sigma$.

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In the morning you put on *socks* ($\rho$), then *shoes* ($\sigma$) $\implies$ $\rho\sigma$.

In the evening you take them off in the *reverse order*: shoes first, then socks $\implies$ $\widehat{\sigma}\,\widehat{\rho}$.

# Reflexivity and Irreflexivity

**Definition 3.13**

A relation $\rho$ on a set $A$ is called *reflexive* if

$$a \ \rho \ a$$

is true for all $a \in A$, i.e., if

$$\mathrm{id} \subseteq \rho.$$

**Definition 3.14**

A relation $\rho$ on a set $A$ is called *irreflexive* if $a \not\rho a$ for all $a \in A$, i.e., if $\rho \cap \mathrm{id} = \varnothing$.

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**Examples on $\{a, b, c\}$:**

$\{(a, a), (a, b)\}$ is *neither*: $(b, b)$ is missing (not reflexive), $(a, a)$ is present (not irreflexive).

$\varnothing$ is irreflexive but not reflexive.

# Symmetry and Antisymmetry

**Definition 3.15**

A relation $\rho$ on a set $A$ is called *symmetric* if

$$a \ \rho \ b \iff b \ \rho \ a$$

is true for all $a, b \in A$, i.e., if $\rho = \widehat{\rho}$.

**Definition 3.16**

A relation $\rho$ on a set $A$ is called *antisymmetric* if

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**Examples on $\{1, 2\}$:**

- $\rho = \{(1, 2), (2, 1)\}$: symmetric, not antisymmetric
- $\sigma = \{(1, 2)\}$: antisymmetric, not symmetric
- $\{(1, 1)\}$: symmetric *and* antisymmetric

# Transitivity

**Definition 3.17**

A relation $\rho$ on a set $A$ is called *transitive* if

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A relation $\rho$ is transitive if and only if $\rho^2 \subseteq \rho$.

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**Example:** $\rho = \varnothing$ and $\rho = \mathrm{id}$ are (vacuously) transitive.

# Transitive Closure

**Definition 3.18**

The *transitive closure* of a relation $\rho$ on a set $A$, denoted $\rho^*$, is

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$$\rho^2 = \{(1,3),(2,4)\}, \quad \rho^3 = \{(1,4)\}, \quad \rho^4 = \varnothing$$

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$\rho^*$ relates two elements iff there is a *path* from one to the other. It is the smallest transitive relation containing $\rho$.

# Equivalence Relations

**Definition 3.19**

An *equivalence relation* is a relation on a set $A$ that is reflexive, symmetric, and transitive.

**Definition 3.20**

For an equivalence relation $\theta$ on a set $A$ and for $a \in A$, the set of elements of $A$ that are equivalent to $a$ is called the *equivalence class* of $a$ and is denoted as $[a]_\theta$:

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Define the relation $\theta$ on $\mathbb{Z}$, such that

$$x \,\theta\, y \iff x \equiv y \pmod 3$$

$$[0]_\theta = 3\mathbb{Z}, \qquad [1]_\theta = 3\mathbb{Z} + 1, \qquad [2]_\theta = 3\mathbb{Z} + 2$$

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"same parity" on $\{1, 2, 3, 4\}$: classes $\{1, 3\}$ and $\{2, 4\}$.

# Intersections of Equivalence Relations

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Check each property:

- $\mathrm{id} \subseteq \sigma$ and $\mathrm{id} \subseteq \theta \Rightarrow \mathrm{id} \subseteq \sigma \cap \theta$
- $(\sigma \cap \theta)\widehat{\ } = \widehat{\sigma} \cap \widehat{\theta} = \sigma \cap \theta$
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What about $\sigma \cup \theta$?

# Partitions and Quotient Sets

**Definition 3.21**

A *partition* of a set $A$ is a set of mutually disjoint subsets of $A$ that cover $A$, i.e., a set $\{S_i \mid i \in \mathcal{I}\}$ of sets $S_i$ (for some index set $\mathcal{I}$) satisfying

$$S_i \cap S_j = \varnothing \quad \text{for } i \neq j \qquad \text{and} \qquad \bigcup_{i \in \mathcal{I}} S_i = A.$$

**Definition 3.22**

The set of equivalence classes of an equivalence relation $\theta$, denoted by

$$A/\theta \stackrel{\text{def}}{=} \{[a]_\theta \mid a \in A\},$$

is called the *quotient set* of $A$ *by* $\theta$, or simply $A$ *modulo* $\theta$, or $A$ **mod** $\theta$.

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**Example:** "same parity" on $\{1, 2, 3, 4\}$:

$$A/\theta = \big\{\{1, 3\}, \{2, 4\}\big\}$$

is a partition of $A$. And $\mathbb{Z}/\equiv_3 = \{3\mathbb{Z},\ 3\mathbb{Z}+1,\ 3\mathbb{Z}+2\}$.

# Equivalence Relations $\Leftrightarrow$ Partitions

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The set $A/\theta$ of equivalence classes of an equivalence relation $\theta$ on $A$ is a partition of $A$.

And the reverse direction holds too: every partition $P$ of $A$ induces the equivalence relation

$$x \; \theta_P \; y : \Longleftrightarrow \; x \text{ and } y \text{ lie in the same block of } P.$$

# Partial Orders

**Definition 3.23**

A *partial order* (or simply an *order relation*) on a set $A$ is a relation that is reflexive, antisymmetric, and transitive. A set $A$ together with a partial order $\preceq$ on $A$ is called a *partially ordered set* (or simply *poset*) and is denoted as $(A; \preceq)$.

**Definition 3.24**

For a poset $(A; \preceq)$, two elements $a$ and $b$ are called *comparable* if $a \preceq b$ or $b \preceq a$; otherwise they are called *incomparable*.

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In $(\mathcal{P}(\{1, 2\}); \subseteq)$ the sets $\{1\}$ and $\{2\}$ are incomparable.

# Total Orders and Covering

**Definition 3.25**

If any two elements of a poset $(A; \preceq)$ are comparable, then $A$ is called *totally ordered* (or *linearly ordered*) by $\preceq$.

**Definition 3.26**

In a poset $(A; \preceq)$ an element $b$ is said to *cover* an element $a$ if $a \prec b$ and there exists no $c$ with $a \prec c$ and $c \prec b$ (i.e., between $a$ and $b$).

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$\leq$ on $\mathbb{N}$ is total; $|$ on $\mathbb{N}$ is not ($2, 3$ incomparable).

In $(\{1, 2, 3, 4, 6, 12\}; |)$: $4$ covers $2$, but $12$ does *not* cover $2$ because $2 \prec 4 \prec 12$.

# Hasse Diagrams

**Definition 3.27**

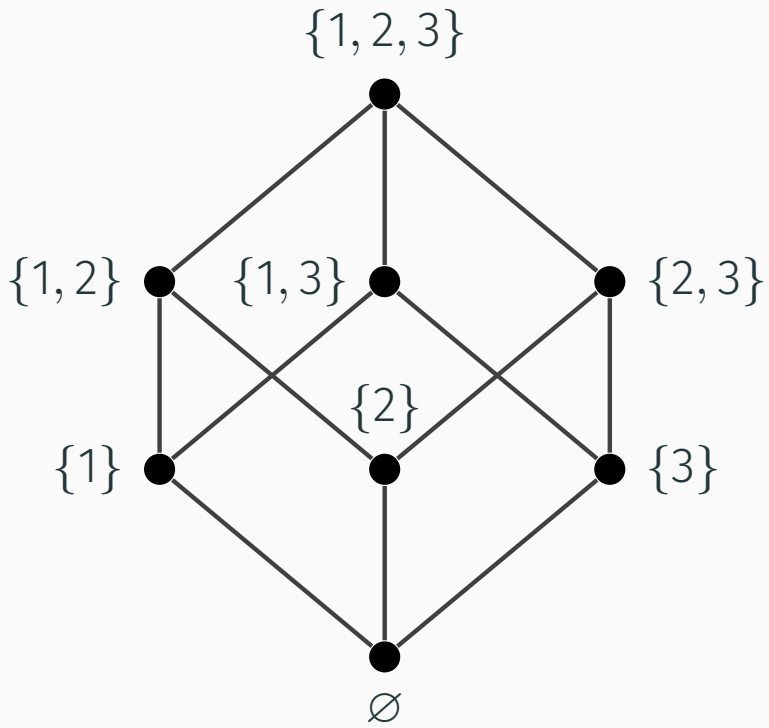
The *Hasse diagram* of a (finite) poset $(A; \preceq)$ is the directed graph whose vertices are labeled with the elements of $A$ and where there is an edge from $a$ to $b$ if and only if $b$ covers $a$.

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Convention: loops, arrow heads and implied (transitive) edges are left out; edges point upwards. $(\mathcal{P}(\{1,2,3\});\ \subseteq)$



# Direct Product of Posets

**Definition 3.28**

The *direct product* of posets $(A; \preceq)$ and $(B; \sqsubseteq)$, denoted $(A; \preceq) \times (B; \sqsubseteq)$, is the set $A \times B$ with the relation $\leq$ (on $A \times B$) defined by

$$(a_1, b_1) \leq (a_2, b_2) \overset{\text{def}}{\iff} a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2.$$

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Example: $(\{0, 1\}; \leq) \times (\{0, 1\}; \leq)$ with product order from Def 3.28.

# Lexicographic Order

**Theorem 3.13**

For given posets $(A; \preceq)$ and $(B; \sqsubseteq)$, the relation $\leq_{\text{lex}}$ defined on $A \times B$ by

$$(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \overset{\text{def}}{\iff} a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

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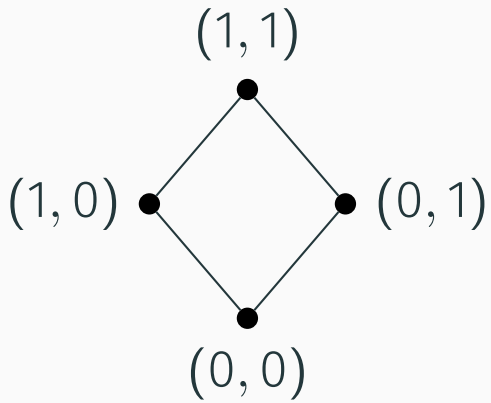
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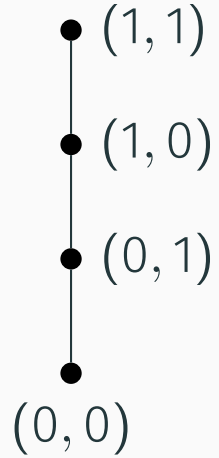
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is a partial order relation.

Same set $\{0, 1\} \times \{0, 1\}$: product order (left) vs. lexicographic order (right)



# Extremal Elements and Bounds

**Definition 3.29**

Let $(A; \preceq)$ be a poset, and let $S \subseteq A$ be some subset of $A$. Then

1. $a \in A$ is a *minimal* (*maximal*) element of $A$ if there exists no $b \in A$ with $b \prec a$ ($b \succ a$).
2. $a \in A$ is the *least* (*greatest*) element of $A$ if $a \preceq b$ ($a \succeq b$) for all $b \in A$.
3. $a \in A$ is a *lower* (*upper*) *bound* of $S$ if $a \preceq b$ ($a \succeq b$) for all $b \in S$.
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In $(\{2, 3, 4, 6, 12\}; |)$: minimal $= \{2, 3\}$, no least element;

maximal $=$ greatest $= 12$.

For $S = \{4, 6\}$: $\mathrm{ub} = \{12\}$, $\mathrm{lb} = \{2\}$, so $\mathrm{lub}(S) = 12$, $\mathrm{glb}(S) = 2$.

# Well-Orders, Meet and Join

**Definition 3.30**

A poset $(A; \preceq)$ is *well-ordered* if it is totally ordered and if every non-empty subset of $A$ has a least element.

**Definition 3.31**

Let $(A; \preceq)$ be a poset. If $a$ and $b$ (i.e., the set $\{a, b\} \subseteq A$) have a greatest lower bound, then it is called the *meet* of $a$ and $b$, often denoted $a \wedge b$. If $a$ and $b$ have a least upper bound, then it is called the *join* of $a$ and $b$, often denoted $a \vee b$.

**Definition 3.32**

A poset $(A; \preceq)$ in which every pair of elements has a meet and a join is called a *lattice*.

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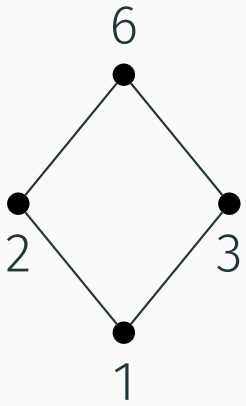
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In $(\mathcal{P}(A); \subseteq)$: meet $= \cap$, join $= \cup$. In $(\mathbb{N}_{>0}; |)$: meet $= \gcd$, join $= \mathrm{lcm}$

# Meet and Join, Visually

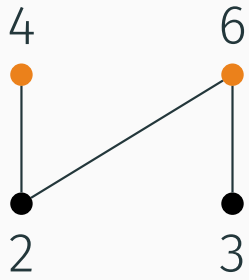
**A lattice:** $(\{1, 2, 3, 6\}; |)$




every pair has meet and join:

$2 \wedge 3 = 1 \qquad 2 \vee 3 = 6$

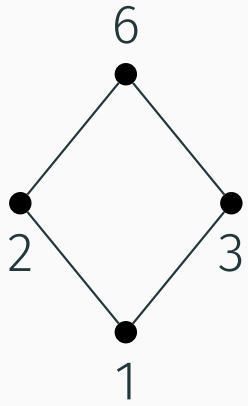
**Not a lattice:** $(\{2, 3, 4, 6\}; |)$




4 and 6 have *no upper bound* in this poset, so $4 \vee 6$ does not exist.

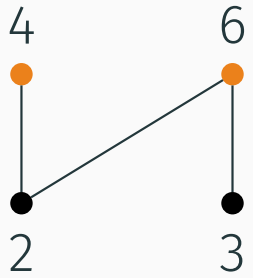
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**Not a lattice:** $(\{2, 3, 4, 6\}; |)$




4 and 6 have *no upper bound* in this poset, so $4 \vee 6$ does not exist.

$2 \wedge 3 = 1 \qquad 2 \vee 3 = 6$

A lattice fails as soon as some pair has no least upper bound

(or no greatest lower bound).

# Exercises

# Union of Equivalence Relations

Prove or disprove: If $\sigma$ and $\rho$ are two equivalence relations (on the same set), then $\sigma \cup \rho$ is also an equivalence relation.

# Union of Partial Orders

Prove or disprove: If $\sigma$ and $\rho$ are two partial order relations (on the same set), then $\sigma \cup \rho$ is also a partial order relation.

# Classify the Relation

A relation $\rho$ on $\mathbb{Z}$ is defined by

$$x \,\rho\, y :\iff xy \geq 0.$$

Which of the properties reflexive, irreflexive, symmetric, antisymmetric, transitive hold?

# Transitive Closure

Compute the transitive closure of

$$\rho = \{(1,2),(2,3),(3,4),(4,2)\}$$

on $A = \{1,2,3,4\}$.

# A Relation on $\mathbb{Z} \times \mathbb{Z}$

A relation $\sigma$ on $\mathbb{Z} \times \mathbb{Z}$ is defined by

$$(a, b)\ \sigma\ (c, d) :\iff a + d = b + c.$$

Prove that $\sigma$ is an equivalence relation and describe its equivalence classes.

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# Exercise 5.7

**Exercise 5.7 Old Exam Question (HS18)**

Let $\rho$ and $\sigma$ be two equivalence relations on a set $A$. Prove that if $\rho \circ \sigma = \sigma \circ \rho$ then $\rho \circ \sigma$ is an equivalence relation. That is, prove that it is reflexive, symmetric and transitive.

(a) reflexive

(b) symmetric

(c) transitive

# Old Exam Task HS 24

**d)** ($\star$) Let $(X, \preceq)$ be a poset. Let $\prec$ be the relation on $X$ defined as follows:

$$a \prec b \iff a \preceq b \ \wedge \ a \neq b.$$

**Prove or disprove** that $\prec$ is transitive.

*(6 Points)*

# Old Exam Task HS 25

**b)** ($\star$) **Prove or disprove** the following statements. For all sets $X \neq \varnothing$ and all relations $\rho$ and $\tau$ on $X$:

1.) $(\rho \cap \tau)^* \subseteq \rho^* \cap \tau^*$,

2.) $(\rho \cap \tau)^* \supseteq \rho^* \cap \tau^*$,

where $\sigma^*$ denotes the transitive closure of $\sigma$. *(5 Points)*

# Questions?

Material: `https://diegostrebel.com/teaching`

